

Key environmental problems

Environmental problems: are issues caused by human activities and cause damage to the environment.

Definition of environmental problems:

It is any change or disturbance to the environment considered to be undesirable or dangerous. It brings down or reduces the capacity of the environment to meet the social and ecological needs.

Environmental problems can be categorized into **2** major factors:

- 1) A **natural factor**: like drought, sea storms, volcanic eruptions etc.
- 2) **anthropogenic factors**: means man made activities: which include deforestation industrialisation, urbanisation actually causing damage to the elements of the Earth which is air water and soil.

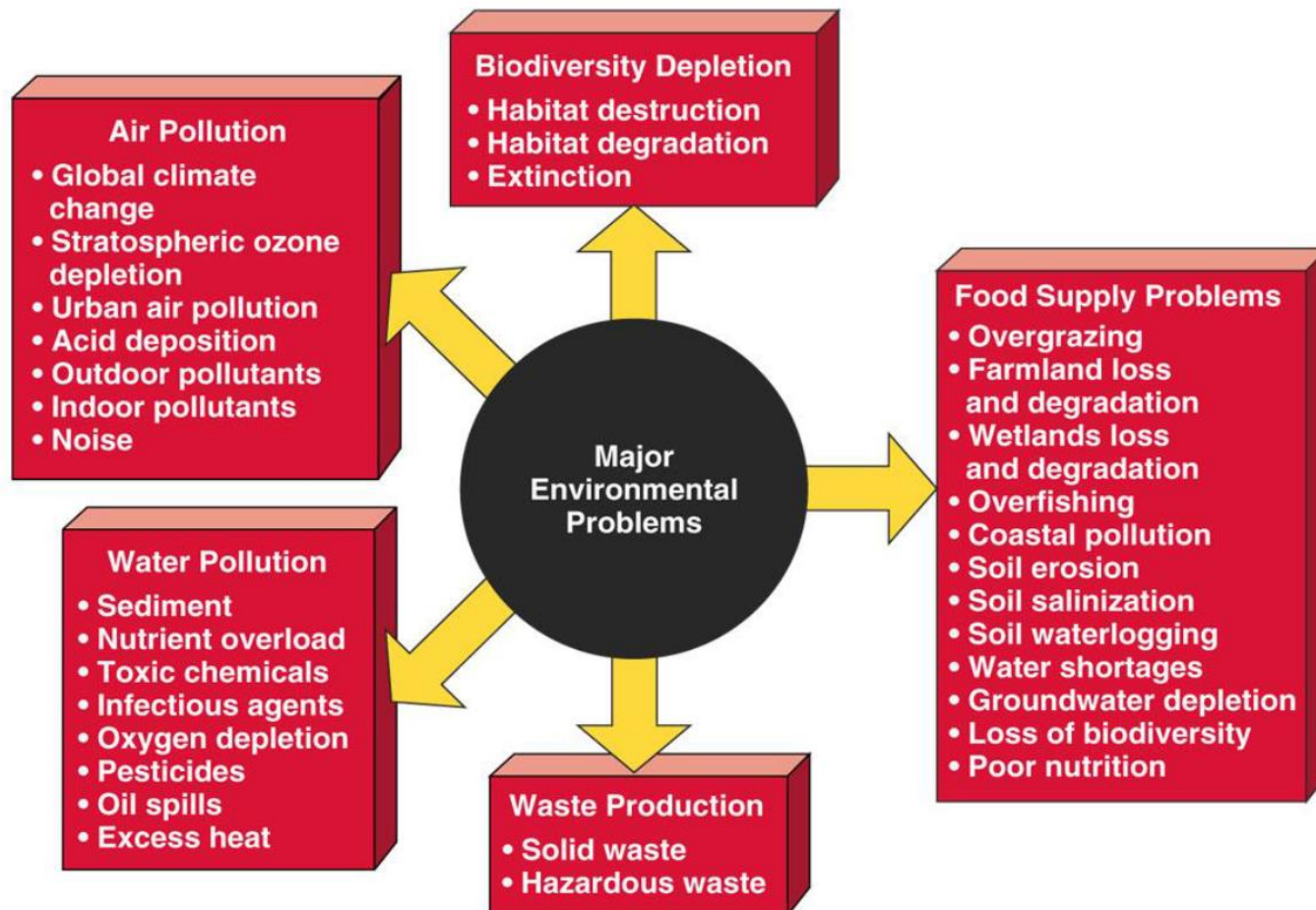
Causes of Environmental Problems: There are several causes: Like population, ozone layer depletion, acid rain, food scarcity, biodiversity loss, waste production, and use of unsustainable resources.

However, **four major causes** of environmental problems are:

- Population growth
- Poverty
- Affluence based on wasteful and unsustainable resource use
- Exclusion of harmful environmental costs from the market prices of goods and services.

The below illustrations brings to us a plethora of environmental problems:

Environmental Problems



One must know about **Natural Resources and Natural Services – key components in nature's sustainability**

- **Natural resources** are materials and energy in nature that are essential or useful to humans. They are often classified as renewable resources (such as air, water, soil, plants, and wind) or non-renewable resources (such as mineral ores, oil, coal).

- **Natural services** are processes in nature, such as purification of air and water and renewal of topsoil, which support life and human economies.

Natural capital is the world's stock of natural resources, which includes geology, soils, air, water and all living organisms. Natural capital assets provide people with a wide range of **free goods and services**, often called ecosystem services, which underpin our economy and society and some of which even make human life possible.

Optimal utilization of natural capital

- In economic terms, *capital refers to money and* other forms of wealth that can support a person, a population, or an economy. It can provide a sustainable income if we use it properly—that is, if we do not spend it too quickly. If we protect capital by careful investment and spending, it can last indefinitely.
- Similarly, **natural capital** can support the earth's diversity of species as long as we use its natural resources and services in a sustainable fashion.

Overuse of natural capital – causes Unsustainability • Over exploitation of non-renewable resources - petroleum - coal - natural gas – minerals

Ecological Footprints: A Model of Unsustainable Use of Resources • Supplying people with renewable resources results in wastes and pollution, and can have an enormous environmental impact this is taken as an **ecological footprint**.

• **Ecological footprint is nothing but the amount of** biologically productive land and water needed to provide the people in a particular country or area with an indefinite supply of renewable resources and to absorb and recycle the wastes and pollution produced by such resource use. It considers only renewable resources though use of non-renewable resources also causes pollution.

- **Population growth:** A population is defined as a group of individuals of the same species living and interbreeding within a given area. The population rely on similar resources and get subjected to similar environmental constraints and depend on the availability of the other members to persist over time. Therefore one must limit the population explosion and use the natural resources wisely.
- **Affluence** comes from the Latin verb affiliate meaning receiving an incoming flood of riches In terms of environmental science, affluence is the abundance of wealth and goods or the consumption of high volumes of goods, those taken from the Earth.
- One must be conscious of how we utilize the natural resources judiciously
- **Affluence Has Harmful Environmental Effects** •The lifestyles of many consumers in more-developed countries and in less-developed countries such as India and China are built upon growing affluence, which results in high levels of consumption and unnecessary waste of resources. Such affluence is based mostly on the assumption—fueled by mass advertising—that buying more and more material goods will bring fulfillment and happiness.
- The harmful environmental effects of affluence are dramatic. The U.S. population is only about one-fourth that of India. But the average American consumes about 30 times as much as the average Indian and 100 times as much as the average person in the world's poorest countries. As a result, the average environmental impact, or ecological footprint per person, in the United States is much larger than the average impact per person in less-developed countries.

- The authors of the book, *Minimalism* stresses us to identify the essentials and eliminate the rest.
- **Poverty:** It is a state of being poor, always in want of money, living space and access to quality air, water, food and basic sanitation.

Environmental Impact: is very challenging. Desperate for short-term survival, some individuals in poverty degrade potentially renewable forests, soils, grasslands, fisheries, and wildlife at an ever-growing demand. Poaching for animal skins for a large price, succumbing to unethical practices for some quick money all these impact the earth. The poor have a tendency to have larger families. This leads to over exploitation of natural resources too! We must act and achieve a world of **ZERO Poverty**—the first sustainable development goal.

- **Excluding cost of environment and natural resources:** Prices Do Not Include the Value of Natural Capital. Companies using resources to provide goods for consumers generally ignore the harmful environmental costs of supplying such goods. Like the leather industry does not worry about the chromium impacts or the ecological index.
- **Consumerism:** is the latest addict affecting the world. The responsible human being must buy only what he needs. Imbibing a culture of green consumerism will benefit us and the earth too!
- **Global warming and climate change:** The green house effect is a natural process that heats up the earth's surface and the atmosphere. Green house gases CO₂, H₂O vapour and CH₄ impact the energy balance of the planet. The amount of heat energy added to the atmosphere by green house effect is controlled by the concentration of greenhouse gases like CO₂, CFCs, nitrous oxide, methane etc. in the earth's atmosphere. As a result of this higher concentration, the green house effect will be enhanced and the earth's climate will become warmer and this is referred to as global warming.

These issues can be addressed only by responsible citizens and the ecological index can be promising only with all our cumulative effort. May we live and bequeath this planet in a better way.